1. Gradient Descent (GD) — the foundation

This is the base idea: update parameters in the direction of negative gradients.

$$\theta = \theta - \eta \nabla J(\theta)$$

- **θ** = parameters
- **η (learning rate)** = step size
- ∇**J(θ)** = gradient of loss

Types of Gradient Descent:

- **Batch GD** → uses full dataset (slow, stable)
- **Stochastic GD (SGD)** → one sample at a time (fast, noisy)
- **Mini-batch GD** → practical standard (balance)

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2. Stochastic Gradient Descent (SGD) (important baseline)

Even though it's a variant, treat it as a separate optimizer.

- Faster updates
- Adds noise → helps escape local minima
- But oscillates a lot

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3. Momentum

Adds “velocity” to smooth updates.

v = \beta v + (1-\beta)\nabla J(\theta), \quad \theta = \theta - \eta v

$$v = \beta v + (1 - \beta)\nabla J(\theta), \quad \theta = \theta - \eta v$$

- Accelerates in consistent direction
- Reduces oscillation
- Think: **rolling ball gaining speed**

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4. AdaGrad (Adaptive Gradient)

Adapts learning rate per parameter.

\theta = \theta - \frac{\eta}{\sqrt{G + \epsilon}} \n

abla J(\theta)

$$\theta = \theta - \frac{\eta}{\sqrt{G+\epsilon}}\nabla J(\theta)$$

- Frequently updated parameters → smaller steps
- Rare features → larger steps
- Problem: learning rate keeps shrinking

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5. RMSProp

Fixes AdaGrad’s shrinking problem.

G = \beta G + (1-\beta)(\nabla J(\theta))^2

$$G = \beta G + (1 - \beta)(\nabla J(\theta))^2$$

- Uses moving average instead of accumulating all gradients
- Works well for **non-stationary problems**

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6. Adam (Adaptive Moment Estimation)

Combines **Momentum + RMSProp**

\theta = \theta - \frac{\eta \cdot m}{\sqrt{v} + \epsilon}

$$G = \beta G + (1 - \beta)(\nabla J(\theta))^2$$

- **m** = first moment (mean)
- **v** = second moment (variance)

Why Adam is popular:

- Fast convergence
- Works well in most problems
- Default choice in practice

Simple Analogy

- GD → walking downhill carefully
- SGD → jumping randomly downhill
- Momentum → rolling ball
- AdaGrad → smart step sizes
- RMSProp → forgetting old history
- Adam → smart + fast + stable